

Mohamed Martini

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Skills: Python, AI/ML (LangChain, LangGraph, TensorFlow, PyTorch), Cloud (GCP - Cloud Run, CloudSQL, Cloud Trace), Data (PostgreSQL, BigQuery, Apache Beam), DevOps (Docker, CI/CD), Web (FastAPI), Flutter.

PROFESSIONAL EXPERIENCE

Founder & Principal Engineer | Lever AI

Full-time · July 2025 - Present

- Founded a consulting and SaaS development LLC specialized in AI-powered, data-driven software solutions.
- **Showcase Project: Guesstimate: Not Trivia!** ([Google Play](#), [App Store](#))
 - Designed and shipped a full-stack social estimation game to Android and iOS, featuring **real-time multiplayer** via Firestore, global leaderboards, and freemium monetization (RevenueCat, Unity Ads).
 - Built an **LLM-powered ETL pipeline** using smart seed selection and pgvector embeddings for **semantic deduplication**, generating 700+ high-quality, diverse, and fact-checked questions at <1 cent per question.
 - Architected real-time, fair question assignment for multi-player rounds using **optimized SQL**.
 - Implemented **end-to-end distributed tracing** with OpenTelemetry, correlating Flutter frontend events to FastAPI backend spans via Cloud Trace and Cloud Logging.
 - Established **GitFlow CI/CD pipelines** with automated testing, and dev/prod building and deployment.
 - Leveraged AI coding agents to accelerate cross-platform Flutter development while maintaining full architectural oversight of the Python backend.

Machine Learning Engineer III | Pison Technology

Full-time · Oct 2022 - July 2025 · 2 yrs 10 mos

- **Promoted from MLE to MLE III** in April 2025.
- Ran 1,200+ TensorFlow experiments for sleep stage classification, improving the baseline F1-score by 15%. Developed two custom loss functions to address class imbalance. Implemented a custom attention layer for feature mining, improving performance while reducing the size of the feature set by an order of magnitude.
- Developed real-time biosignal gesture detection models. Wrote Cythonized extension modules, and vectorized bottleneck functions, significantly improving the speed of online prediction and offline modeling.
- Led a "Hackaweeek" project with two colleagues to implement a state-of-the-art [domain-invariant classification network with adversarial training](#) in PyTorch, following the approach outlined in its original manuscript.
- Led the adoption of BigQuery as a data warehouse solution to streamline the handling of analytics data.
- Deployed and monitored streaming and batch ML models using Google Model Registry and Cloud Functions.
- Wrote a scalable ETL pipeline using Apache Beam which brought down the compute time of 56 GB of data from 2 days to 15 minutes at a cost of 25 cents.
- Standardized the Python codebase, CI/CD workflows, and dependency management, enabling seamless collaboration between colleagues across operating systems.
- Maintained cross-team communication to ensure consistency and optimal integration of ML solutions.

Research Assistant | UNIVERSITY OF MASSACHUSETTS LOWELL

Jan 2021 – June 2022

- Implemented **Reinforcement Learning** agents for cooperative multi-agent Search and Rescue simulations.
- Implemented **object detection** training and inference pipelines. Used Generative AI for data augmentation.

EDUCATION

M.S. Computer Engineering	(4.0 GPA)	UMass Lowell	2022
B.S. Electrical Engineering	(3.8 GPA)	UMass Lowell	2021
B.S. Biology	(3.5 GPA)	UMass Lowell	2019